

Plug-and-Play Deep Image Prior for Snapshot Optical Coherence Tomography

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Abstract

We demonstrate a high-throughput computation-efficient snapshot coherence tomographic imaging method by combining interferometric coding and compressive sampling. The depth distribution of a three-dimensional (3D) object is encoded into the spectrum of a light field, using the principle of optical coherence tomography (OCT), i.e., through a Michelson interferometer, to generate an intermediate (x, y, λ) data-cube that encodes the raw (x, y, z) data of the object [1]. We then sample the spectral data using a well-established compressive spectral imaging technique, called the coded aperture snapshot spectral imaging (CASSI) [2], which yields a compressed 2D measurement that captures the whole 3D tomographic information of the object. Finally, A self-supervised neural network is developed that integrates deep image priors (DIP) into the plug-and-play (PnP) regime for tomographic reconstruction from the single 2D measurement. Such integration of OCT and CASSI leads to a physically simple and computationally efficient system.

Principle of snapshot OCT

Fig. 1 illustrates the principle of our method. We first convert the 3D (x, y, z) distribution of the object into a spatial-spectral (x, y, λ) datacube through a Michelson interferometer (indicated with the red dashed box in Fig. 1) [3]. A wide-field broadband light beam is used to illuminate both a reference mirror and the object. The reflected beams from the mirror and the object are then combined to form the required (x, y, λ) interference signal, which encodes the 3D object. This z to λ conversion adopts the idea from OCT.

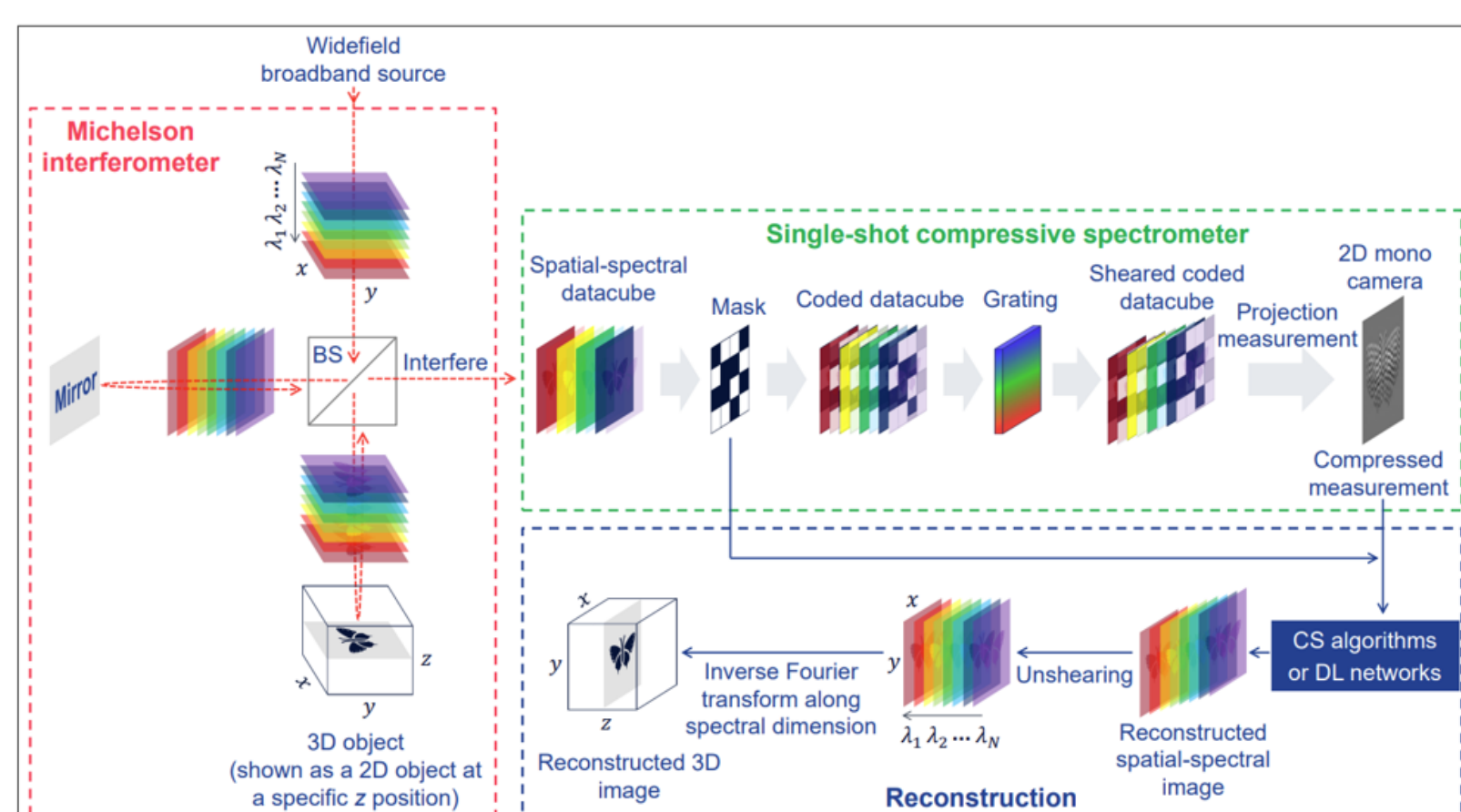


Figure 1: Schematic diagram of snapshot OCT

The interference intensity I for wavelength λ can therefore be expressed as an integral over the z -axis:

$$I(\lambda) = \int I_r + I_s(z) + 2\sqrt{I_r I_s(z)} \cos\left(2\pi \frac{2z}{\lambda}\right) dz, \quad (1)$$

By subtracting the direct current (DC) terms I_r and $I_s(z)$ from Eq. (1), we obtain the desired interference signal I_{AC} which will be used to calculate the depth distribution:

$$I_{AC}(k) \propto \int \sqrt{I_s(z)} (e^{i2zk} + e^{-i2zk}) dz, i = \sqrt{-1} \quad (2)$$

By applying an inverse Fourier transform to $I_{AC}(k)$, we obtain an image of object's depth distribution as $\sqrt{I_s(z)} (e^{i2zk} + e^{-i2zk})$.

PnP-DIP for snapshot OCT reconstruction

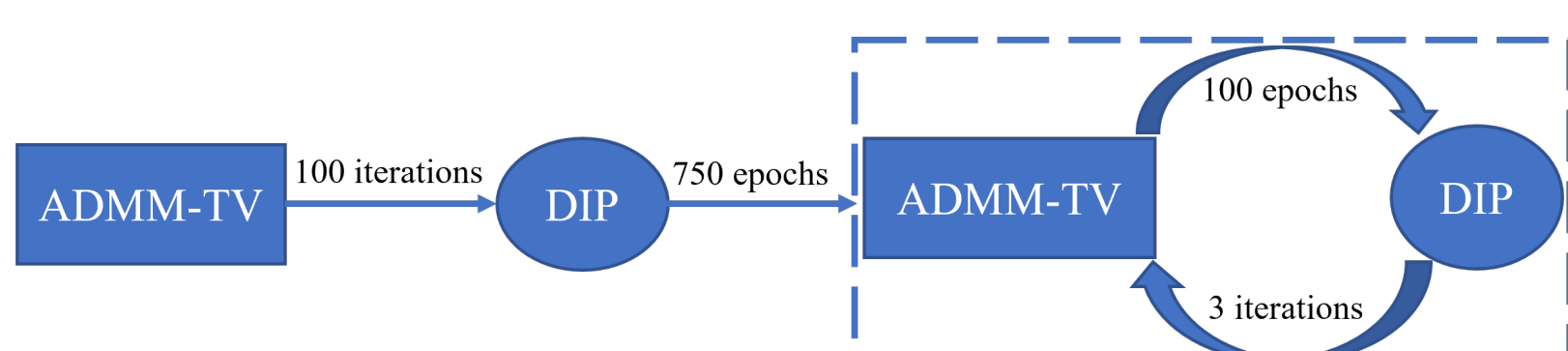


Figure 2: The schematic of the PnP-DIP method

As for snapshot OCT reconstruction, it is not easy to train an end-to-end network for snapshot OCT reconstruction as the generation of 3D depth dataset is difficult, the training process is also time and resource

consuming. So we propose a self-supervised method that integrates deep image priors (DIP)[4] into the plug-and-play (PnP) regime [5] for the 3-D tomographic reconstruction. we use ADMM-TV [6] for the first 100 times iterations to get the preliminary result and then DIP iterates 750 epochs for further optimization. What follows is the alternate iteration of ADMM-TV and DIP.

Experiment and Results

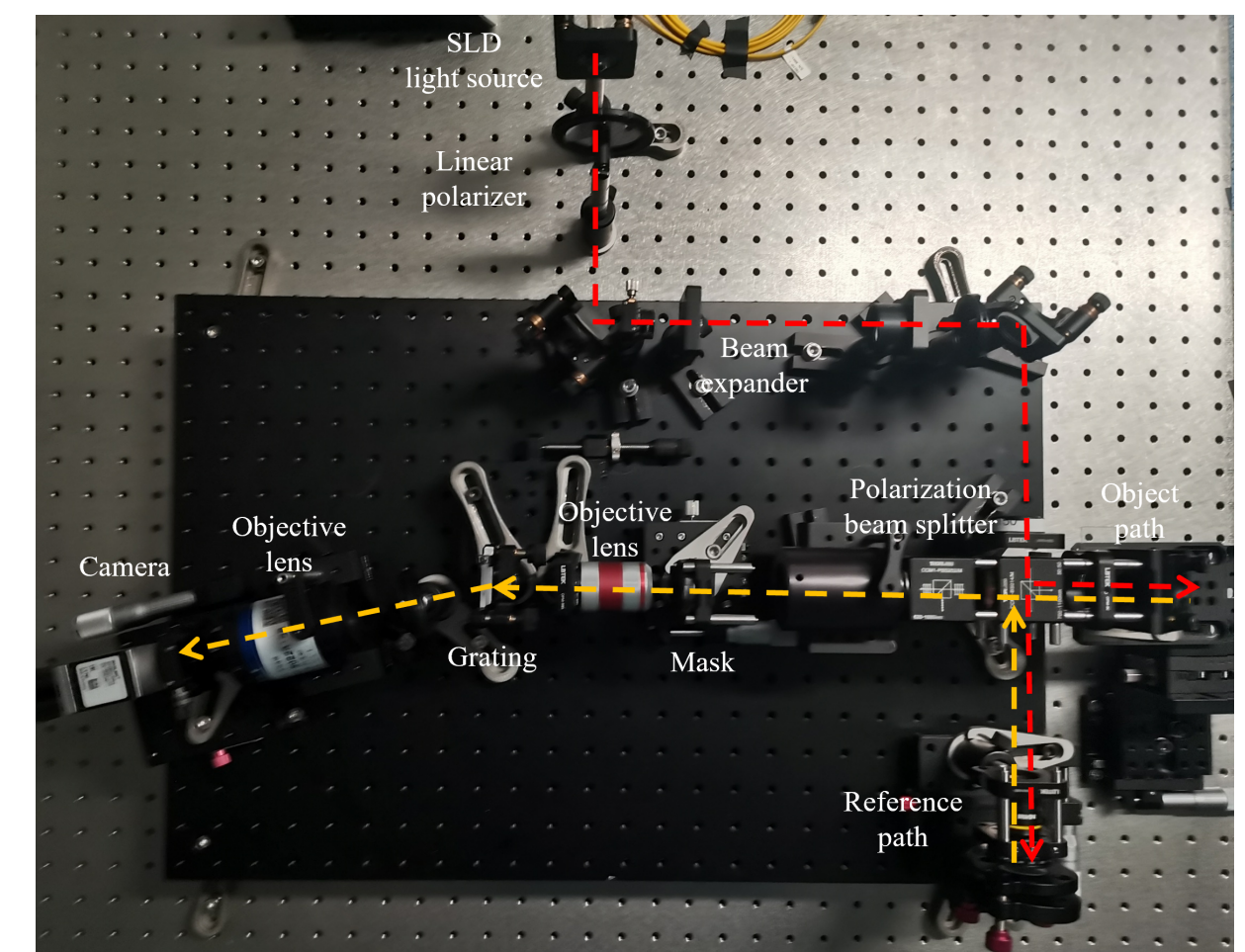


Figure 3: The experimental setup

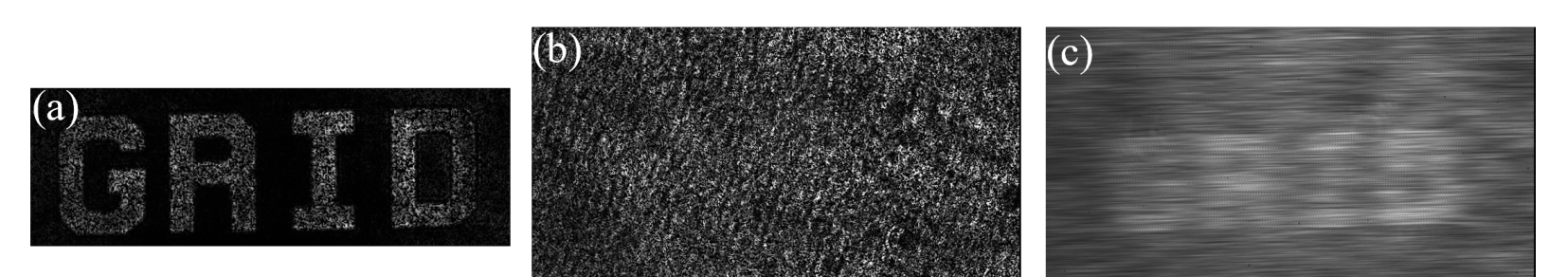


Figure 4: The captured experimental data

Fig. 3 presents the real experimental setup, and the real captured data is presented in Fig. 4, where (a) presents the resolution target with characters 'GRID', (b) is the captured mask and (c) is the compressed 2D measurement that contains the whole 3D tomographic information of the object. With the captured mask and measurement, the PnP-DIP is implemented for reconstruction. A set of reconstructed results is presented as follows, we can find that the proposed PnP-DIP further denoises the results and improves the reconstruction quality.

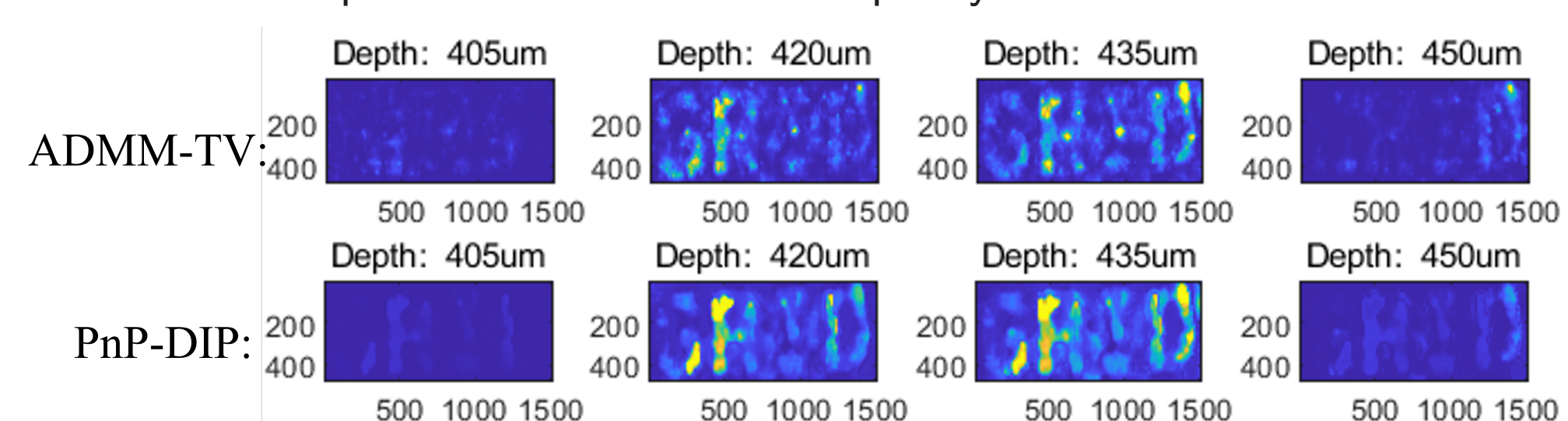


Figure 5: The reconstructed results using ADMM-TV and PnP-DIP

References

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